Session Objectives

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- Describe the formation of normal heartbeat
- Explain how the cardiac impulse is conducted throughout the heart and its physiological importance
- Interpret conduction velocity in different parts of the heart and its determinants
- Analyze the sympathetic effect on heart rate, electrical conduction and contractility
- Compare and contrast the parasympathetic effect with sympathetic effect on heart rate, electrical conduction and contractility

Source: Course Syllabus

Cardiac impulse generation (Pacemaking)

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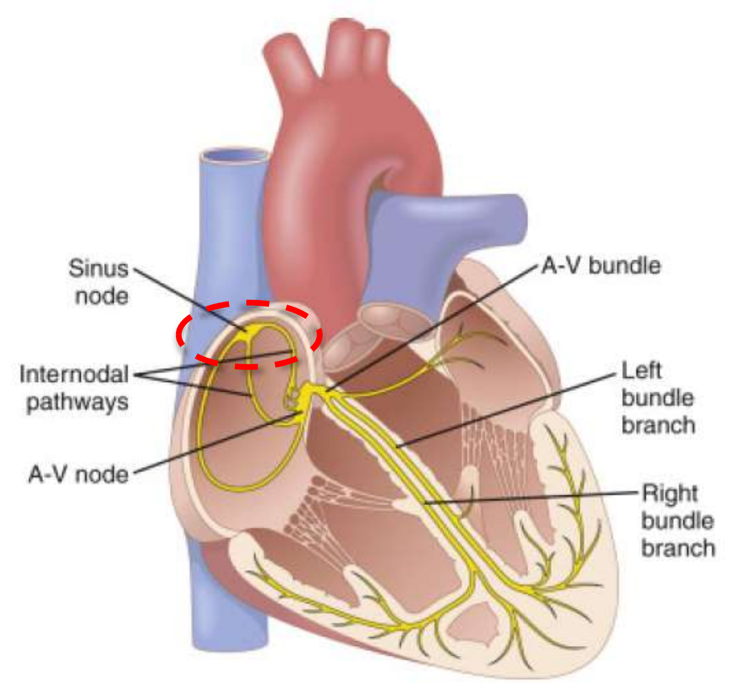
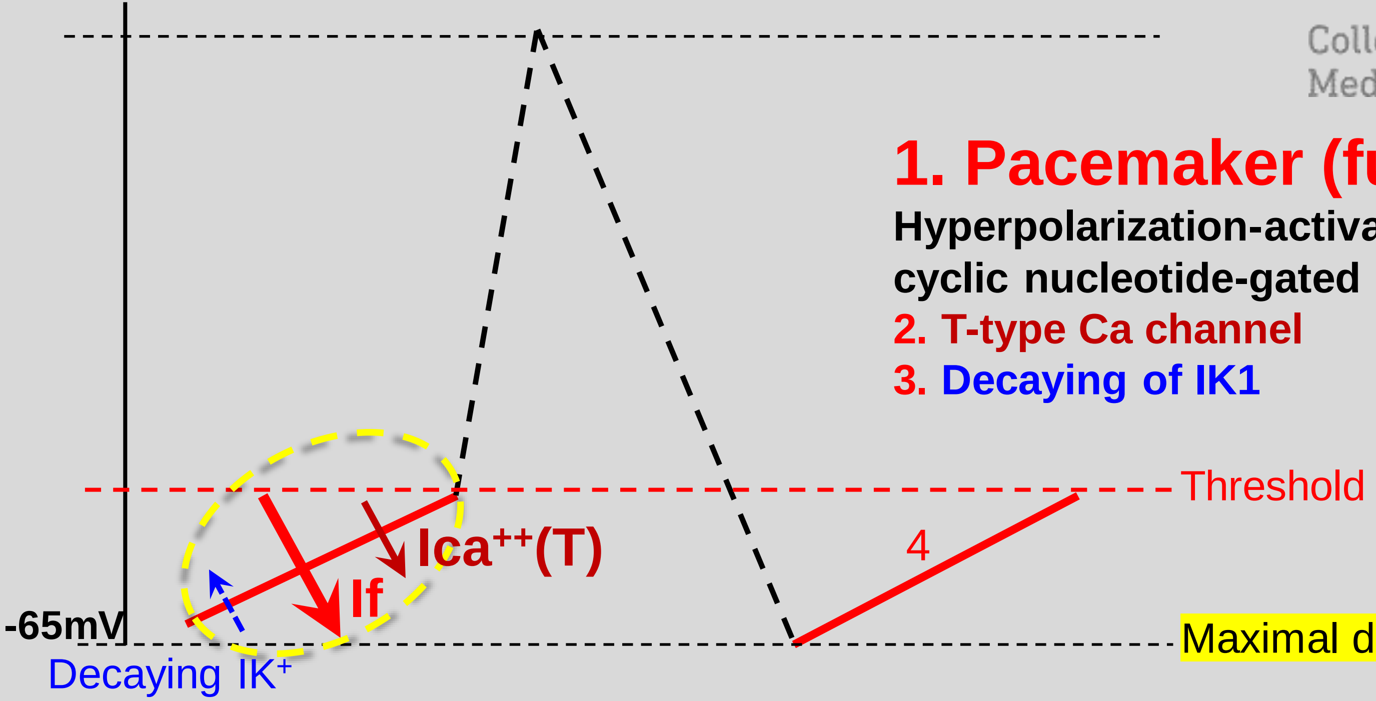


Normal heartbeat is initiated inside the sinoatrial (or sinus) node in the form of slow response action potentials

Guyton and Hall Textbook of Medical Physiology, e13, Figure 10-1

Mechanisms of pacemaking activity

Phase 4 spontaneous depolarization



1. Pacemaker (funny) channel

Hyperpolarization-activated cyclic nucleotide-gated (HCN) channel

2. T-type Ca channel

3. Decaying of IK1

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- Pacemaker current: Hyperpolarization-activated non-selective cation current (Na^+, K^+) or funny current (If).
- It is different from the fast Na^+ channel seen in fast response fibers

Modulation of pacemaker activity (2)

Maximum diastolic potential level

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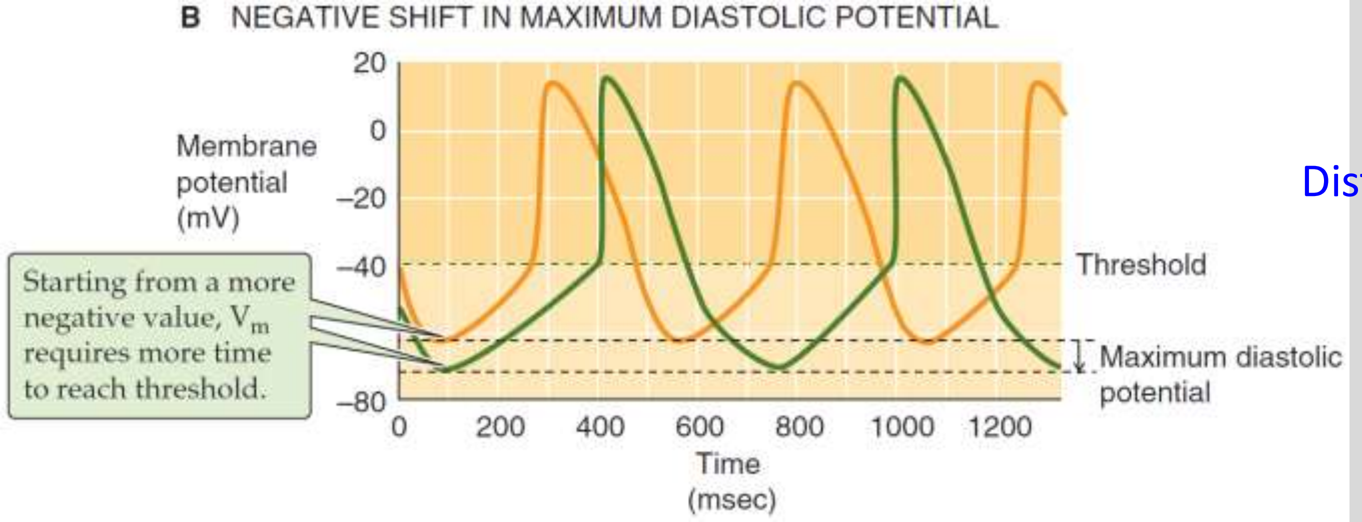

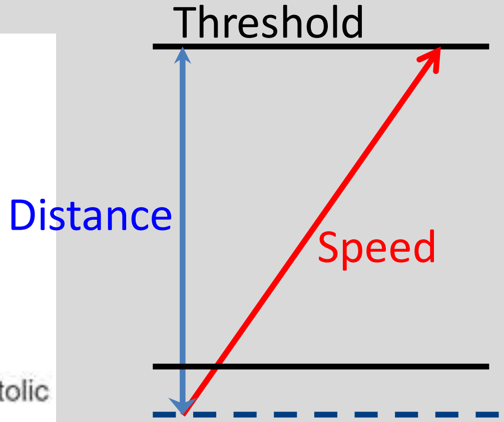

Resting potential (Maximum diastolic potential)

Boron & Boulpaep, Medical Physiology UP Ed, PP492

Modulation of pacemaker activity (3)

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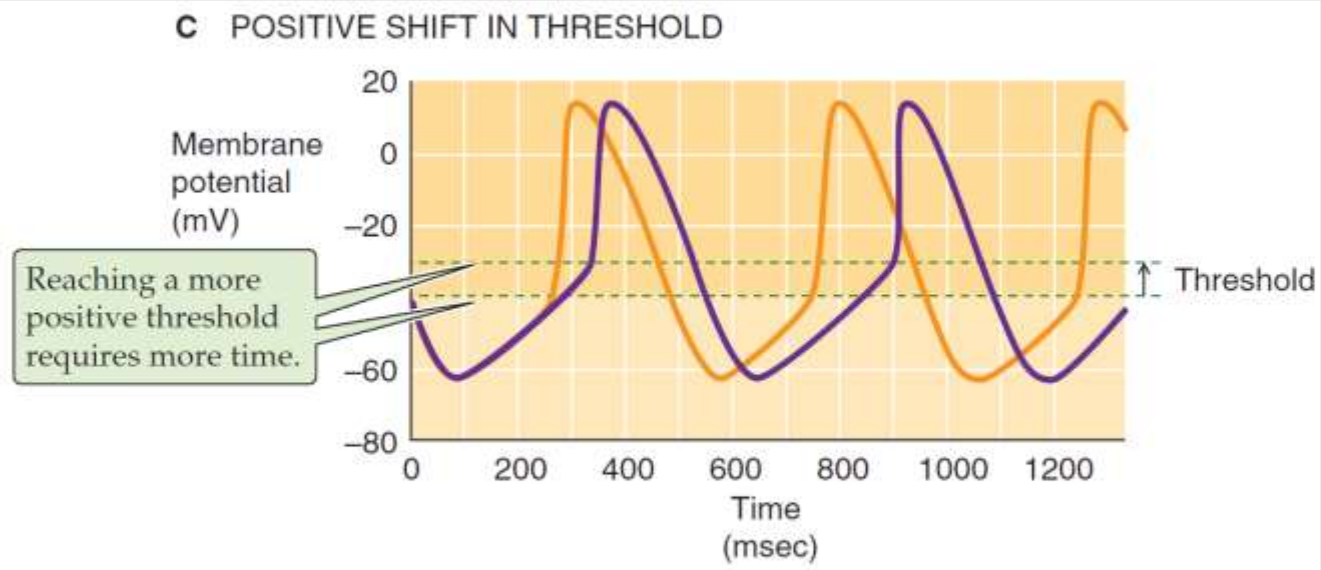
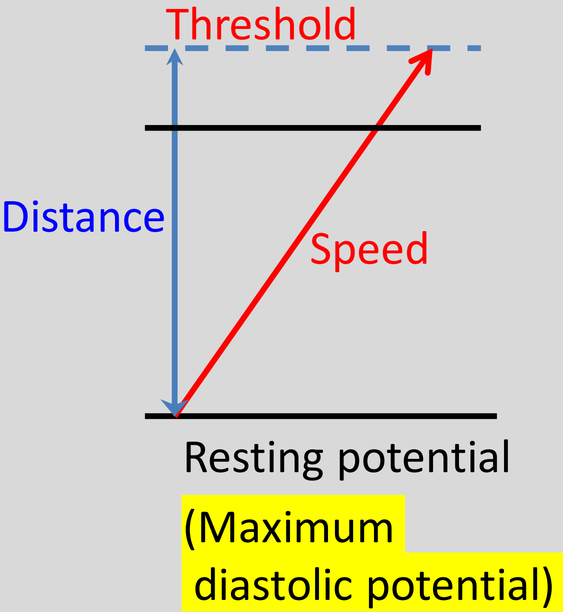
Threshold level





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Cardiac pacemaker hierarchy

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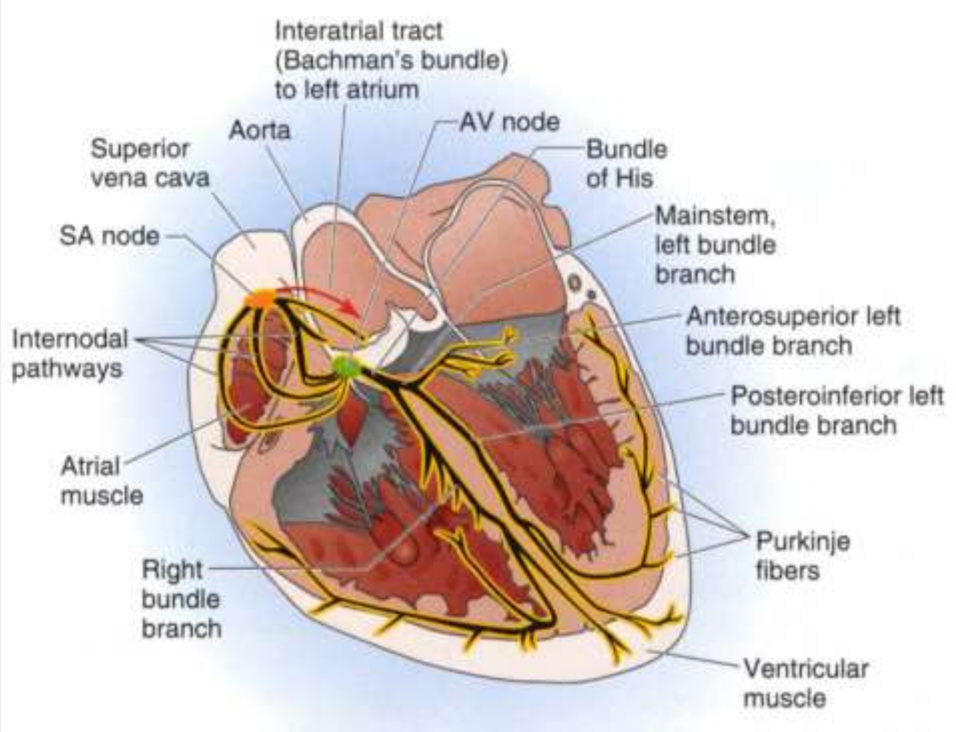

1. **Sinus node:** 60-100bpm (primary)
2. **Atrioventricular node:** 40-60bpm (secondary)
3. **His-Purkinje system:** 20-40bpm (ventricular escape beats)

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Cardiac Impulse Propagation

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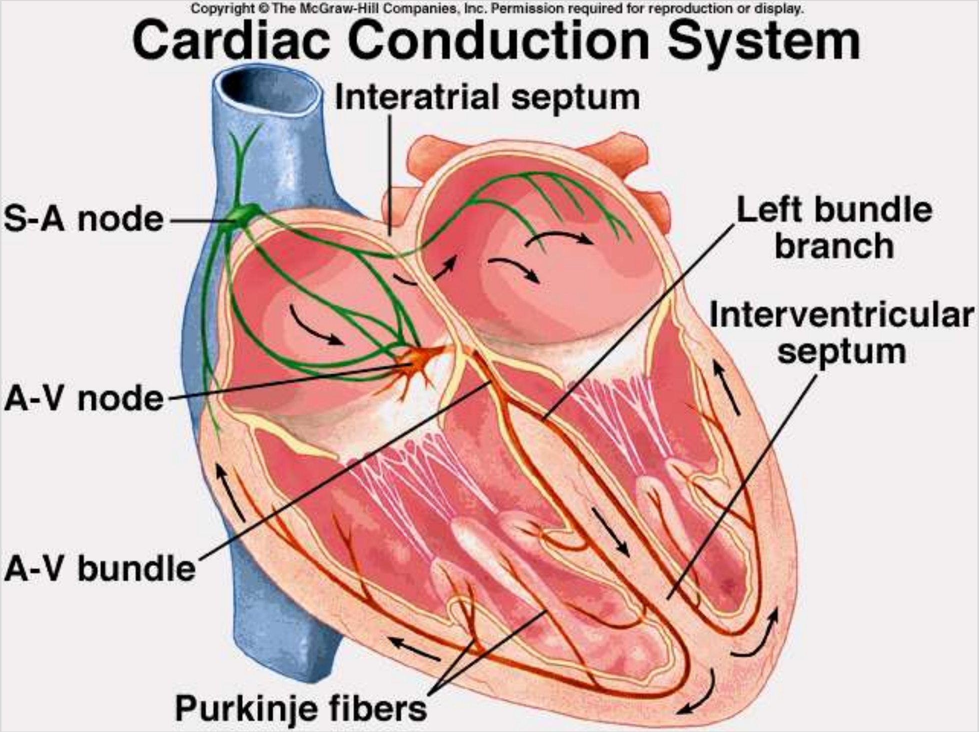

DAILYEM - WordPress.com

Heart beat (AP) initiated in

SA node → Atria (R-L) →

AV node → His bundle →

Bundle branches (R-L) →

Purkinje fibers → Ventricles

The propagation sequence determines: 1. ECG waveforms

2. Also affects cardiac function

Sinoatrial node (primary pacemaker)

(SA node or sinus node)

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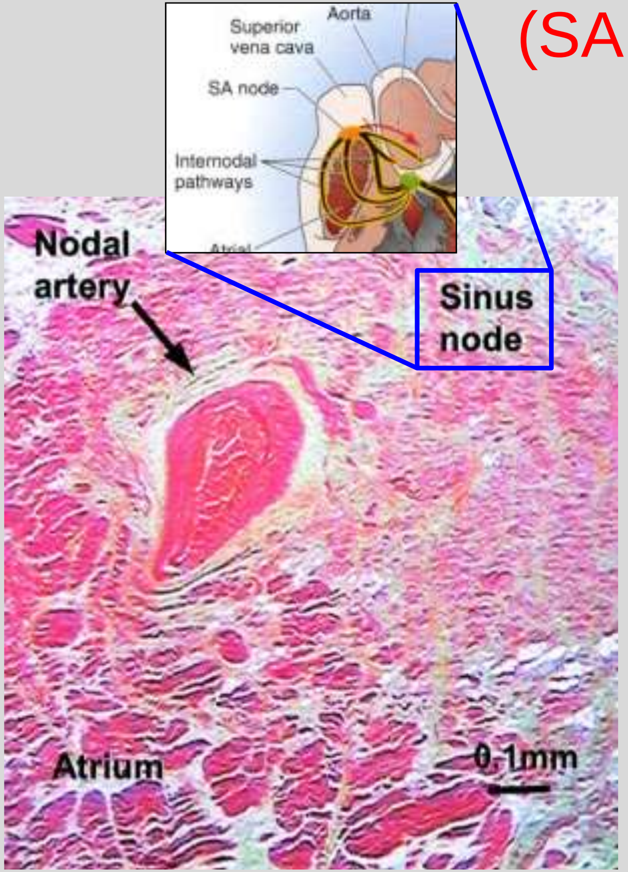

1. Location: Superior vena cava (SVC) & right atrium (RA) junction, or High-RA clinically
    - Center: nodal or p-cells
    - Periphery: transitional cells between nodal and atrial fibers

2. Gap Junction: connects adjacent cardiac cells, allowing ions to pass between cells (sinus node: cx45)

3. Heart rate: 60-100bpm (average 72bpm)
    - >100bpm=tachycardia
    - <60bpm=bradycardia

Boron & Boulpaep Medical Physiology UP Ed PP484, Figure 20-1
Todor N. Mazgalev et al. Circulation. 2001;103:2660-2667

Electrical propagation within atria

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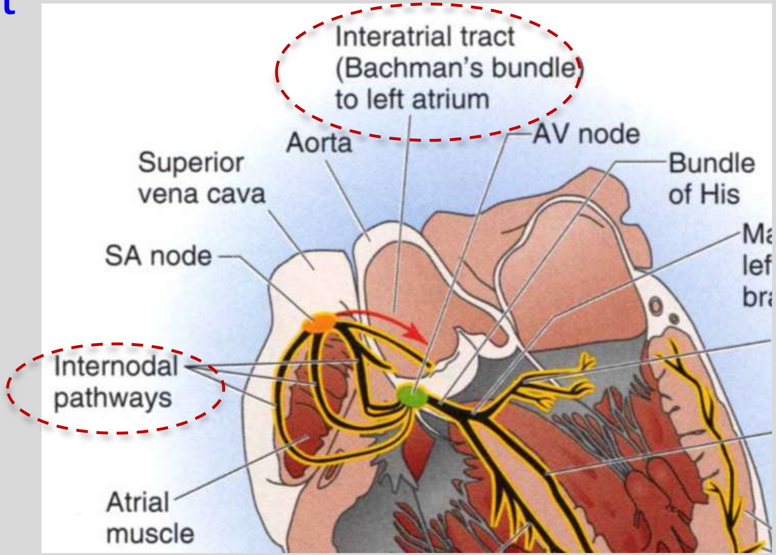
1. From SA node (located at high RA)
2. Going down (top to bottom in RA) through 3 internodal pathways
3. Also from RA to LA via Bachman's bundle (located at the roof of the atria)

- General direction: Top→bottom, right→left

(atrial depolarization creates P wave on ECG)

- P wave axis: downward and leftward

Note: Internodal pathways and interatrial tract are muscle bundles (preferential pathways), not specialized cardiac myocytes.



Boron & Boulpaep Medical Physiology UP Ed PP484, Figure 20-1

Longitudinal vs Transverse Conduction

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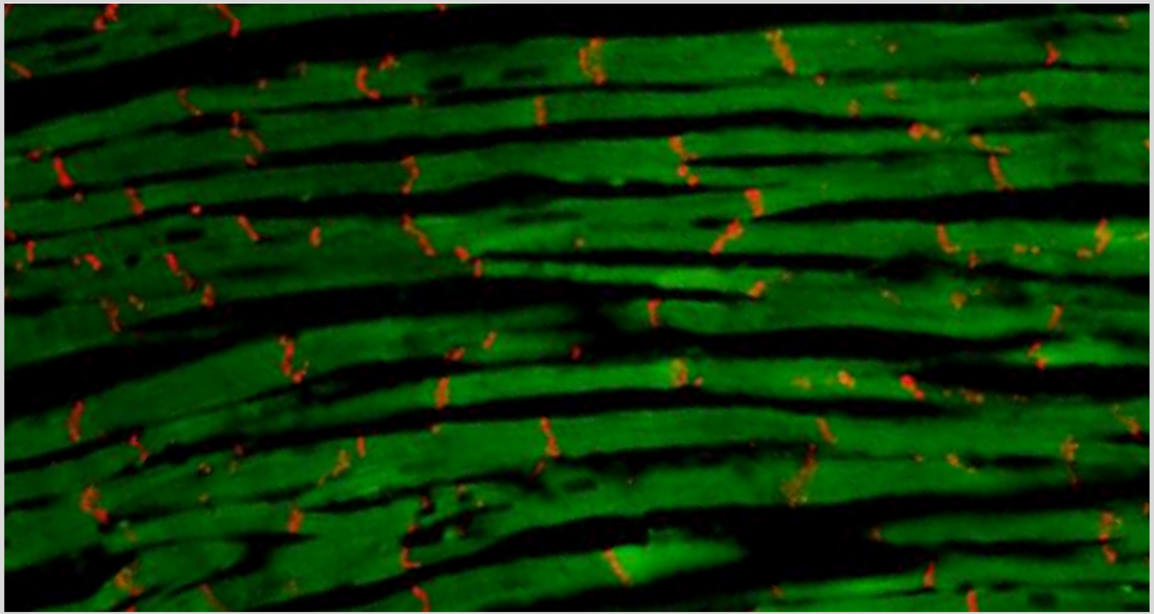
Faster longitudinal conduction (less cells to travel, more gap junctions)



Fast conduction along fiber orientation

Slower transverse conduction (more cells to travel, less gap junctions)

Cx43 (red) in ventricular tissue (our own data)

Gap Junctions

Made of different connexins

(Cxs 40, 43, 45 etc)

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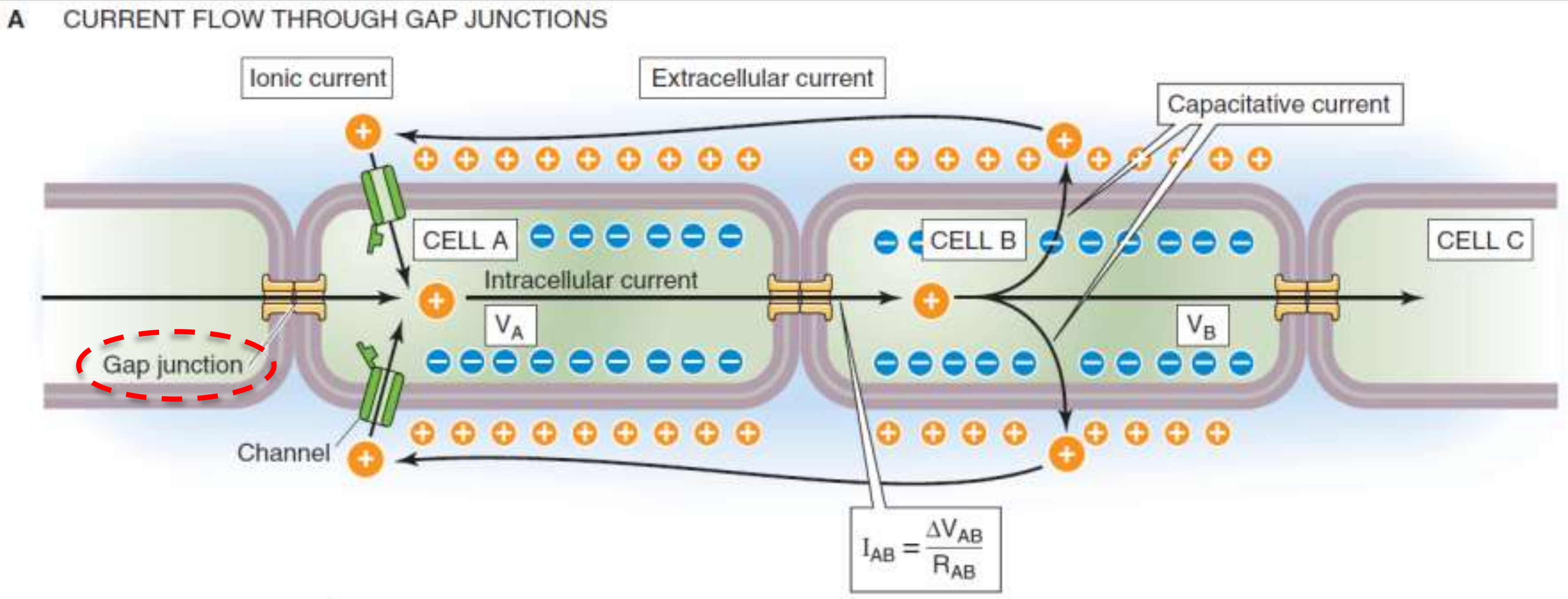

Boron Medical Physiology Figure 21-3A.

Electrical propagation within atria (atrial depolarization creates P wave)

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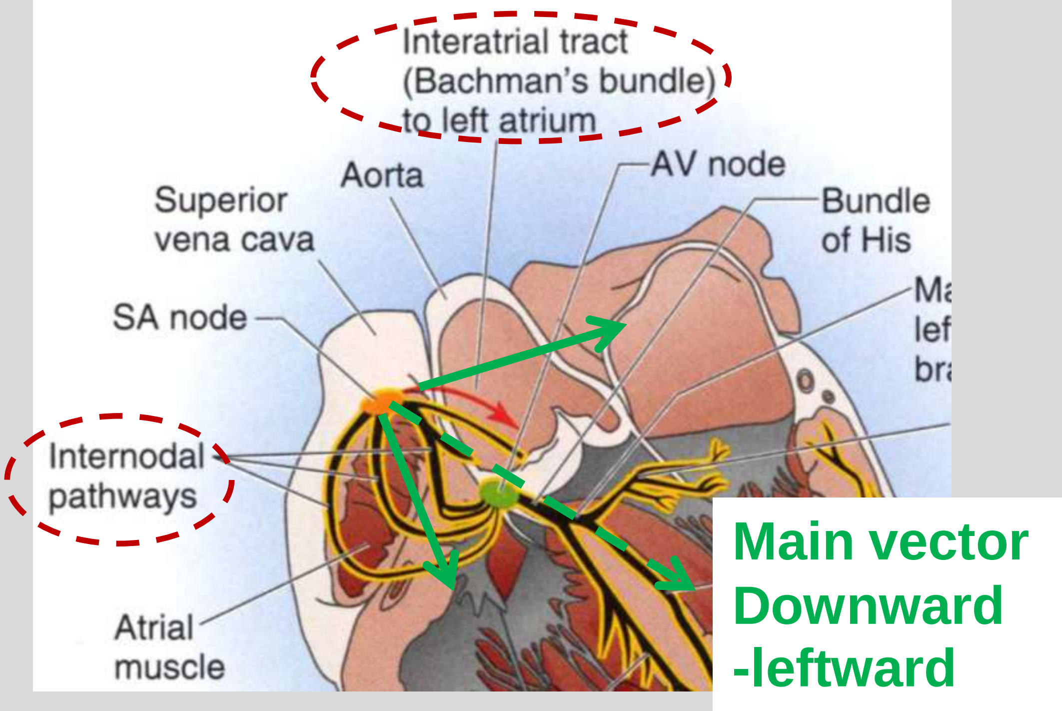

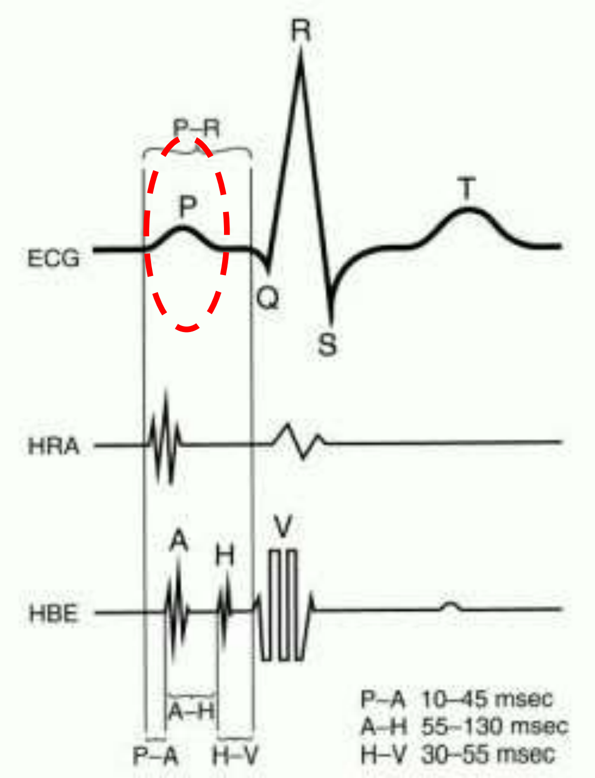

Boron & Boulpaep Medical Physiology UP Ed PP484, Figure 20-1

Atrial excitation and contraction

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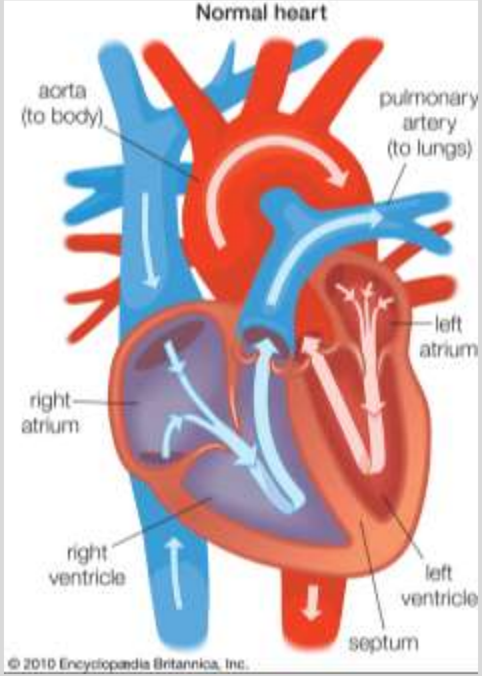
- Excitation of both atria is from top→bottom direction
- Helps to move blood down into the ventricles (lower chambers)



Triangle of Koch (viewed from RA)

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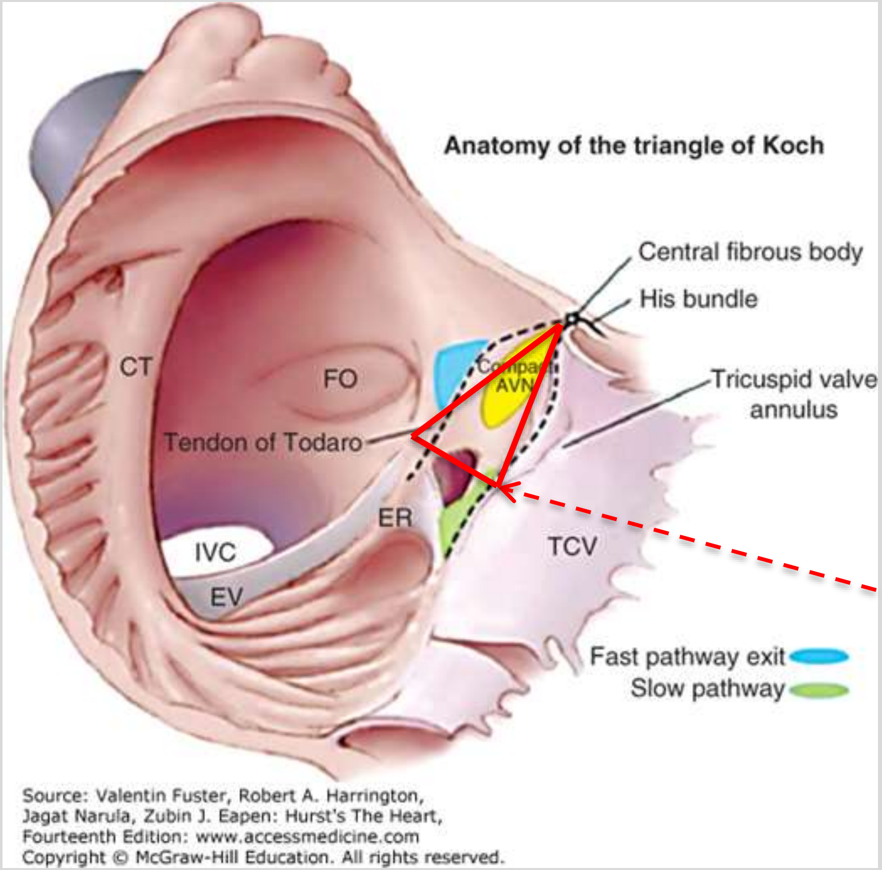

Ostium of coronary sinus

Fig. 78-9 in Hurst's The Heart 14e

His-Purkinje system & ECG

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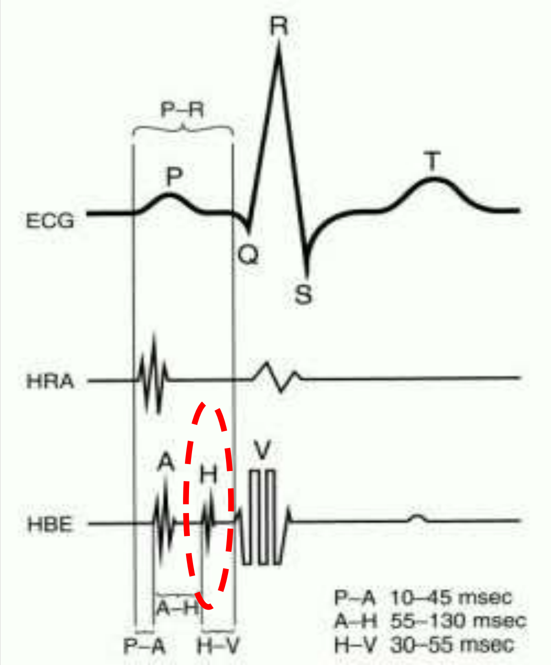
Bundle of His: not shown on ECG,
can be recorded with intracardiac catheter, as shown in the figure (H, His electrogram)

Bundle branches: not shown on ECG
Bundle branch block can be diagnosed with ECG, because it alters normal ventricular activation sequence



Purkinje fibers have the fastest conduction velocity

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Mechanisms

1. Fast response AP (fast INa^+)
   high dv/dt (maximal speed of depolarization) & amplitude
2. Largest cell size
3. High conductance connexin (Cx 43)

Normal excitation is important for ventricular Function

Contraction starts endocardially (inside wall) & from apex toward base (aorta)-pump blood up into aorta

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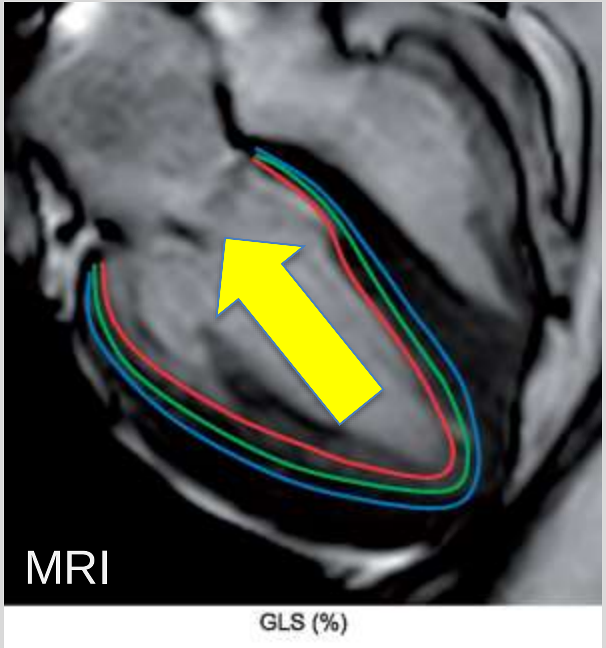
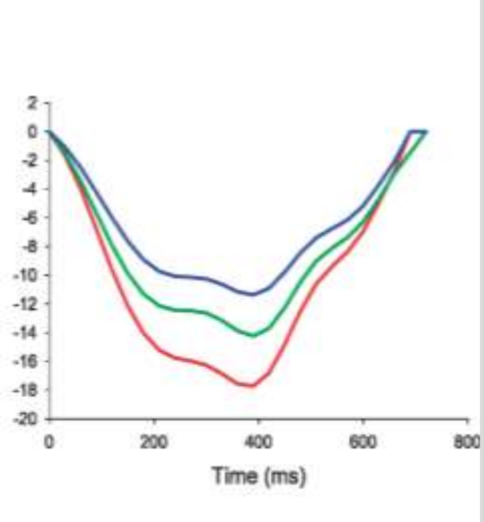


Eur Heart J Cardiovasc Imaging 2019;20(6): 605–619



Contraction starts inside and spread to outside

Curr Cardiol Rev. 2017 May; 13(2): 118–129

Ventricular repolarization sequence

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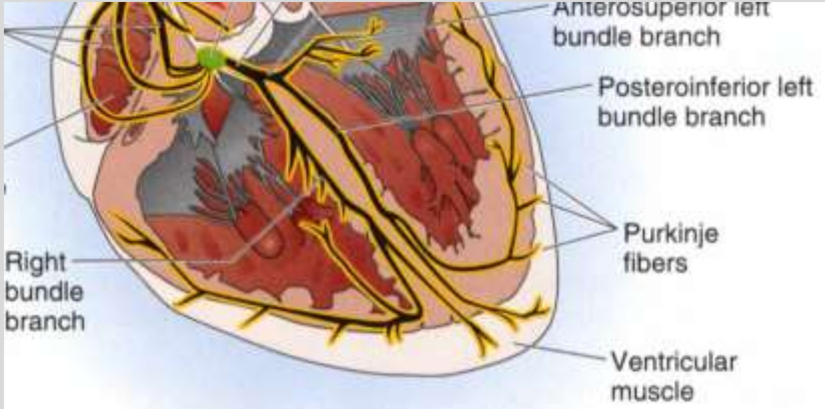
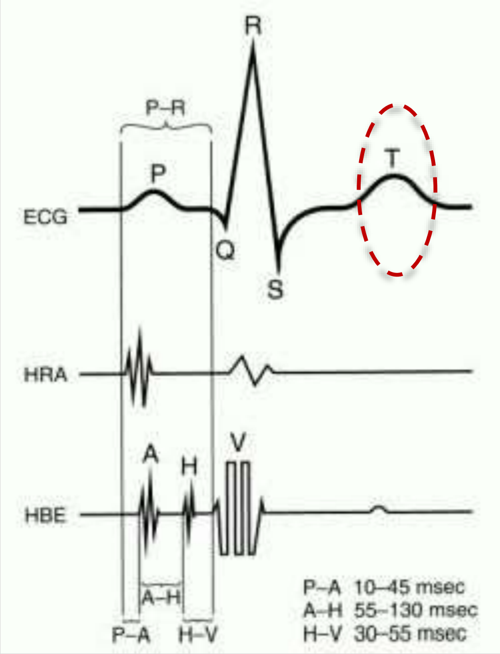
1. Epicardium → endocardium
   (opposite to depolarization)
2. Apex → base
   (same as depolarization)
3. Creates T wave on ECG
   (normally follows QRS polarity)





Boron & Boulpaep Medical Physiology UP Ed PP484, Figure 20-1

Functional importance

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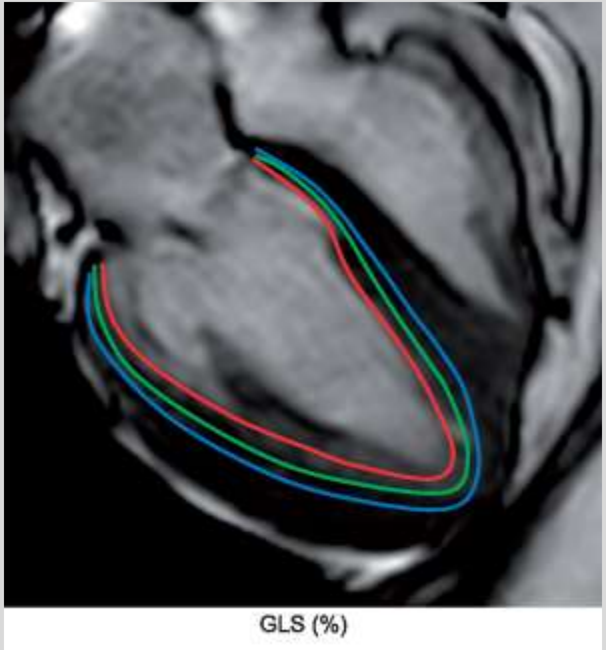
Relaxation starts at apex, epicardially (outside wall) to optimize ventricular relaxation



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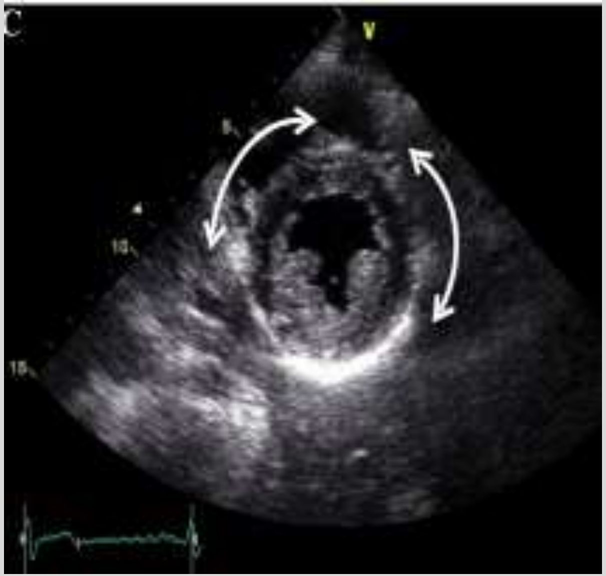
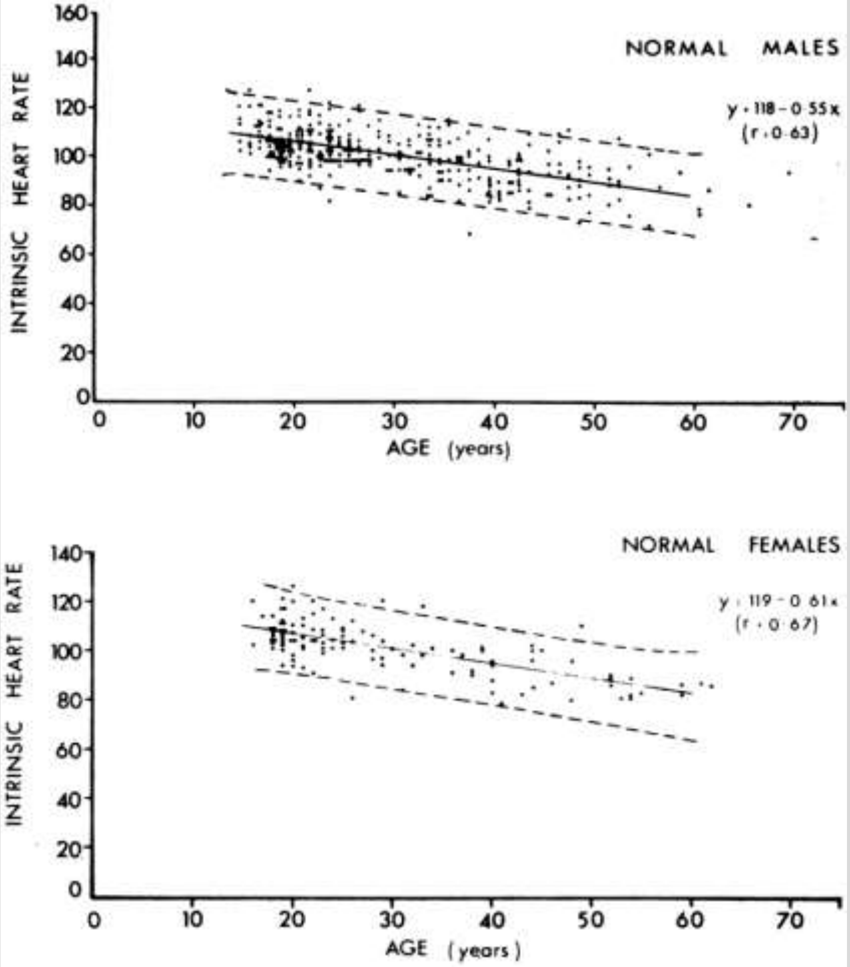


Curr Cardiol Rev. 2017 May; 13(2): 118–129

Concept of intrinsic heart rate



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Without autonomic influence

Block both vagal and sympathetic effects

Cardiovasc Res 1970;4:160–167